

Comparing Transformer and Baseline Models for Multi-Horizon Stock Return Prediction with Technical and Event Features

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Abstract

We study daily multi-horizon equity return prediction on the Dow 30 with horizons $h \in \{1, 5, 21\}$ using technical indicators and earnings features. We compare models across sequence (GRU, LSTM), transformer (TFT), and tabular (Ridge, Random Forest, XGBoost) classes under a strict temporal split. In this run, news features were planned but unavailable; all main experiments therefore use Technical+Earnings inputs (any reported “+News” ablation uses the same inputs as Tech+Earnings). At $h = 1$, LSTM attains the lowest RMSE (0.01622), 0.8–1.5% below the strongest tabular baselines and essentially tied with GRU (0.17% difference). At $h = 5$, Ridge achieves the best RMSE (0.03651), 4–8% lower than tree ensembles and $\sim 37\%$ lower than deep sequence/transformer models. At $h = 21$, LSTM’s RMSE (0.05084) is within 0.03–0.24% of GRU/TFT yet 31–38% below tabular baselines; it also yields the best DA (55.0%), beating GRU/TFT by ~ 1.1 pp and tabular models by 0.8–3.4 pp. All R^2 values are ~ 0 or slightly negative because the benchmark is a zero-return predictor; this is typical for daily returns and makes directional accuracy the more discriminative metric here. Regime slices at $h = 21$ show GRU lowest RMSE in the 2022 bear and LSTM lowest RMSE in the 2023–24 rally, while DA is highest around earnings windows (61.4% vs. 53.9% otherwise). We also report a simple long-short equity curve for XGBoost and TFT at $h = 21$. Overall, simple sequence models are competitive on daily panels, while tabular baselines remain strong on RMSE at intermediate horizons.

1 Introduction

Multi-horizon equity prediction remains challenging given noisy targets, regime dependence, and the potential for only small predictive signals. We present a clear, apples-to-apples comparison across model classes on the Dow 30 with realistic event sensitivity, evaluating how different modeling approaches perform under consistent data splits and feature sets.

Contributions.

- *Robust experimental design:* Strict, reproducible temporal

split (2016–2019 train, 2020 val, 2021–2024 test) and Dow 30 constituents fixed as of 2018-01-01 to mitigate survivorship bias.

- *Model class comparison:* Compare sequence/transformer vs. tabular models on identical technical + earnings feature sets; (news features were planned but unavailable in this run).
- *Comprehensive evaluation:* Evaluate RMSE, R^2 , and DA across $h \in \{1, 5, 21\}$, plus detailed regime (bear vs. rally) and earnings window slices.
- *Insights:* Identify where each model class wins (e.g. bear vs. rally regimes; around earnings windows) and discuss why simpler models can rival more complex ones on daily financial data.

Hypotheses

H1 (Model class advantage). Given multi-horizon forecasting with technical and earnings features on a pooled Dow 30 panel, the Temporal Fusion Transformer (TFT) will achieve lower RMSE and higher DA than strong baselines (LSTM, GRU, XGBoost, Random Forest, Ridge) on the 21-day horizon.

H2 (Event features add value). Adding earnings/event features to technical indicators will improve DA (and not worsen RMSE) relative to technical-only inputs at the 21-day horizon.

H3 (Regime dependence). Model rankings differ by market regime: sequence models (LSTM/GRU) will outperform tabular and transformer models during trending rally periods, while performance gaps will narrow or reverse during bear markets.

Decision criteria. We evaluate each hypothesis using the held-out test set (2021–2024) across three random seeds:

- *H1:* TFT beats the best non-TFT model at $h=21$ by (i) a statistically significant Diebold–Mariano test on squared error ($p < 0.05$) and/or (ii) a McNemar or paired test on directional hits ($p < 0.05$), with effect sizes of at least ≥ 2 pp in DA or $\geq 5\%$ relative improvement in RMSE.

- *H2*: Tech+Earnings vs Tech-only shows DA $\uparrow$ (≥ 2 pp) and RMSE $\downarrow$ (95% CI overlap) for at least one strong model (e.g., TFT or XGBoost) at $h=21$.
- *H3*: Best-in-class models differ across regimes (2022 bear vs 2023–24 rally), verified by per-regime leaderboards and non-overlapping bootstrap CIs.

Outcome transparency. Because News features were unavailable in this run, H1 specifically tests TFT on *technical+earnings* inputs; we treat H2 as a clean test of event-feature value and defer a News-related hypothesis to future work.

2 Related Work

Classical statistical models and tree ensembles are widely used in financial prediction, especially for cross-sectional signals. In particular, simple linear regressions and factor models have a long history, and more recently gradient boosting and random forests have been applied to capture nonlinear effects in asset returns [1, 2]. Gu et al. [1] find that machine learning methods (notably boosted trees and neural networks) substantially improve cross-sectional stock return prediction relative to linear benchmarks, though the gains in traditional metrics can be modest. Deep sequence models such as LSTM and GRU networks [3, 4] have also shown strong performance on time-series tasks and have been explored for financial time series forecasting, due to their ability to capture temporal dependencies and long-term patterns. Recently, Transformer-based architectures have gained popularity for time-series forecasting in finance and beyond, thanks to their attention mechanism that can capture long-range interactions [5]. The Temporal Fusion Transformer (TFT) introduced by Lim et al. [5] targets interpretable multi-horizon forecasting via specialized attention over static metadata, known future inputs, and observed historical covariates.

Event-driven and regime-specific effects are also well-documented in financial markets. For example, the post-earnings announcement drift is a phenomenon where stock returns exhibit predictable behavior following an earnings surprise, indicating that earnings and event features can carry short-term predictive value. Such effects often depend on market regimes and careful alignment to avoid leakage (e.g., ensuring that after-hours earnings announcements are only used for subsequent day predictions). Our work touches on these aspects by including earnings surprise features and examining performance in different market regimes. Compared to prior studies, we directly contrast a transformer model with both deep recurrent models and simpler tabular models on the same dataset, shedding light on whether the added complexity of attention-based models is justified for daily return prediction.

3 Data & Experimental Design

Universe. The Dow 30 stock universe is frozen as of 2018-01-01 (see Appendix A for constituents) to mitigate survivorship

bias.

Splits. We use a strict temporal split: Train set spans 2016-01-01 to 2019-12-31; Validation is 2020-01-01 to 2020-12-31; Test is 2021-01-01 to 2024-12-31. Models are selected and tuned on the validation set, then evaluated on the held-out test.

Targets. The prediction targets are log returns $r_{t \rightarrow t+h} = \ln(P_{t+h}) - \ln(P_t)$ for horizons $h \in \{1, 5, 21\}$ trading days (approximately 1 day, 1 week, 1 month). Directional accuracy (DA) is computed as the fraction of correct sign predictions: $DA = \Pr[\text{sign}(\hat{r}) = \text{sign}(r)]$.

Features. We include a range of technical indicators (standard momentum signals, moving averages, volatility measures, oscillator indicators like RSI, etc.) as well as earnings and event features. Earnings features include the earnings surprise percentage, an indicator for whether earnings were announced after market close or before open (with values aligned to the next trading day), binary event flags for the days of earnings announcements, and days-to-next-earnings countdown features. News-based features were planned but are unavailable in this run¹, so our comparison focuses on technical and earnings inputs; accordingly, all main experiments (EXP-A) are Tech+Earnings.

Leakage controls. We take careful steps to avoid look-ahead bias. All after-market data (such as an earnings result announced after 4pm) is shifted to be only available from $t + 1$ onward. We use purged rolling splits with a 5-day embargo between training, validation, and test periods to ensure no overlap or label leakage. All feature scaling (e.g. standardization) is fit on the train set only, and applied to later sets to prevent information bleed.

Global training. All models are trained on a pooled panel that stacks all 30 tickers (i.e., each training sample is a (date, ticker) pair). The models therefore learn a single set of parameters across all series, leveraging cross-sectional information in addition to time-series patterns. This global modeling approach assumes that different stocks share some common dynamics that the model can learn.

Ablations. Ablations at $h = 21$ compare Tech-only against Tech+Earnings. Because news ingestion failed in this run, the Tech+Earnings+News variant is unavailable and collapses to Tech+Earnings; we therefore report B1 (Tech) and B2 (Tech+Earnings) only (see Table 2). Representative models: XGBoost (tabular) and TFT (transformer). GRU/LSTM use the standard modality inputs and do not participate in the ablation table.

Regime/event slices. We further break down performance on the test set by market regime and by earnings event windows. Specifically, we define a *bear market* slice corresponding to 2022-01 to 2022-10, and a *rally* slice covering 2023-01 to 2024-12. Although a 2020 COVID-crash slice exists in our definitions, it lies outside the 2021–2024 test span and is omitted from this analysis. We also define an *earnings window* slice comprising the periods within ± 3 trading days of each earnings announcement for a stock, versus non-earnings periods. These slices allow us to analyze whether certain models excel during particular market conditions or around informational events.

¹News features were planned but unavailable in this run.

Reproducibility. We fix random seeds $\{42, 43, 44\}$ for all training runs and report averaged results. Sequence models use a lookback window of 60 trading days. Batch size is 256 (per the comparison pipeline). Deep models train up to 10 epochs with early stopping on Val-2020 (validation monitored for patience-based stopping); final metrics are computed once on Test (2021–2024). Baseline hyperparameters are selected on Val-2020 at $h = 21$ by RMSE, then baselines are retrained on Train+Val and evaluated once on Test (hyperparameters fixed across horizons). See Appendix B for details.

4 Models

Tabular models. We evaluate three representative non-sequential learning approaches that treat each (ticker, date) sample as an independent feature vector: (1) *Ridge* regression (linear model with L_2 regularization), (2) *Random Forest* (an ensemble of decision trees), and (3) *XGBoost* (gradient boosted decision trees [2]). These models ignore sequential order (beyond any features we derive like momentum), providing a baseline for how well static relationships in the engineered features can predict future returns. Each tabular model is trained separately for each horizon h with a single-target output.

Sequence models. We use a Gated Recurrent Unit (*GRU*) network [4] and a Long Short-Term Memory (*LSTM*) network [3] to capture temporal dependencies. Each sequence model processes the past 60 trading days of features for a given stock and produces a prediction for $r_{t \rightarrow t+h}$. We include one output head per model (separately trained for each horizon). These recurrent models are well-suited to time-series data and can, in principle, learn patterns like trends or mean-reversion from the sequential input. We apply dropout and L_2 weight decay to RNN models to mitigate overfitting.

Transformer model. We employ the Temporal Fusion Transformer (*TFT*) of Lim et al. [5] to represent the transformer class. The TFT is an attention-based architecture specifically designed for interpretable multi-horizon forecasting. It uses a combination of gated linear units and multi-head self-attention to incorporate different types of covariates: static attributes, known future inputs (e.g., calendar information), and observed historical inputs. Unlike the sequence models which have separate single-horizon versions, the TFT is trained to predict all horizons $h \in \{1, 5, 21\}$ simultaneously via a multi-horizon output layer. Given our relatively small daily panel dataset, we kept the TFT model capacity modest (e.g., limiting the number of attention heads and layers, and using substantial dropout) to prevent overfitting. In this run, the TFT is instantiated with no external news inputs (news_dim=0).

Hyperparameters for all models were selected based on the 2020 validation set (primarily tuning on the $h = 21$ case) and are detailed in Appendix B. Notably, the TFT required careful tuning of learning rate and regularization strength due to its complexity, and simpler architectures like the RNNs were less prone to overfit given the data volume. TFT is trained with a multi-horizon head, whereas RNNs/tabular baselines are trained separately per horizon; we favor realism in how these models

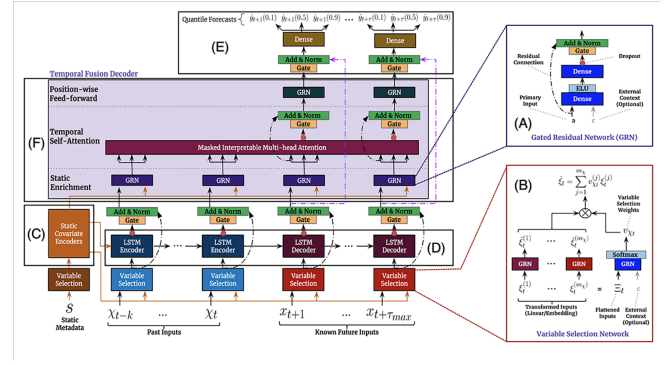


Figure 1: TFT architecture showing how static metadata, observed historical inputs, and known future inputs enter the model through different pathways before fusion via multi-head attention.



Figure 2: Directional accuracy by model and horizon on the test set (Dow 30; 2016–2024 split). This highlights the improvement in predictive direction accuracy for longer horizons: sequence models (LSTM/GRU) particularly gain an edge at $h = 21$.

are typically used over identical objective forms.

5 Results

Table 1 summarizes the performance of each model class on the test set across the three prediction horizons. All metrics are averaged over the 2021–2024 test period for the Dow 30 stocks (with three seeds averaged). We report RMSE (lower is better), R^2 (relative to a zero-return baseline), and DA (higher is better).

Overall, we find that no model achieves a substantially positive R^2 ; in fact, all R^2 values are near zero or slightly negative, underscoring the difficulty of short-term stock return prediction (essentially, none of the models beat a naive baseline in terms of MSE). Nonetheless, there are clear differences in RMSE and DA that highlight each model class’s strengths. At $h = 1$ (one-day ahead), the LSTM attains the lowest RMSE (0.01622), narrowly outperforming other models on error, while Random Forest achieves the highest DA (51.9

Ridge’s win at $h = 5$ suggests much of the weekly signal is approximately linear in our engineered features.

Table 1: Main results on the test set (Dow 30; $h \in \{1, 5, 21\}$; train/val/test split 2016–2024). Each cell shows mean $\pm$ std over 3 random seeds. R^2 is calculated against a zero-return baseline on the test set. Metrics computed from our aggregated results file (Table1_EXP-A.csv).

Model	Horizon h	RMSE	R^2	DA
GRU	1	0.016249 $\pm$ 0.000047	-0.006072	0.500920 $\pm$ 0.024931
GRU	5	0.057883 $\pm$ 0.000123	-0.002885	0.542105 $\pm$ 0.036308
GRU	21	0.050854 $\pm$ 0.000059	-0.002185	0.539142 $\pm$ 0.033861
LSTM	1	0.016222 $\pm$ 0.000025	-0.002677	0.516688 $\pm$ 0.015768
LSTM	5	0.057849 $\pm$ 0.000048	-0.001694	0.554207 $\pm$ 0.000000
LSTM	21	0.050840 $\pm$ 0.000070	-0.001629	0.550429 $\pm$ 0.000000
Random Forest	1	0.016387 $\pm$ 0.000007	-0.023148	0.519454 $\pm$ 0.000290
Random Forest	5	0.039717 $\pm$ 0.000175	-0.206505	0.531379 $\pm$ 0.000636
Random Forest	21	0.081805 $\pm$ 0.000151	-0.248432	0.542956 $\pm$ 0.001095
Ridge	1	0.016346 $\pm$ 0.000000	-0.018058	0.500254 $\pm$ 0.000000
Ridge	5	0.036514 $\pm$ 0.000000	-0.019737	0.511699 $\pm$ 0.000000
Ridge	21	0.074214 $\pm$ 0.000000	-0.027501	0.528230 $\pm$ 0.000000
TFT	1	0.016453 $\pm$ 0.000313	-0.031783	0.489379 $\pm$ 0.019597
TFT	5	0.057907 $\pm$ 0.000108	-0.003690	0.550134 $\pm$ 0.012030
TFT	21	0.050963 $\pm$ 0.000225	-0.006502	0.538742 $\pm$ 0.029354
XGBoost	1	0.016464 $\pm$ 0.000000	-0.032850	0.515368 $\pm$ 0.000000
XGBoost	5	0.038217 $\pm$ 0.000000	-0.117070	0.508611 $\pm$ 0.000000
XGBoost	21	0.081344 $\pm$ 0.000000	-0.234415	0.516713 $\pm$ 0.000000

RMSE in log-return units; R^2 vs zero-return baseline; DA = hit rate on sign.

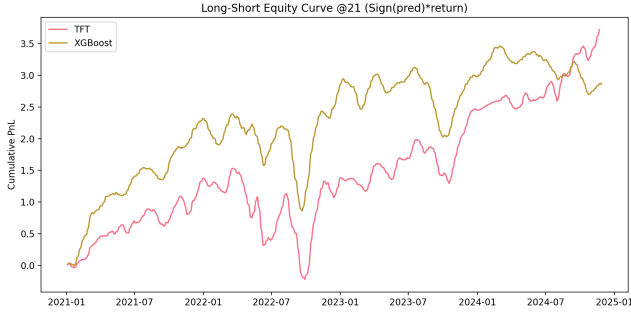


Figure 3: Daily long-short strategy PnL at $h = 21$ using $\text{sign}(\hat{r}) \times r$ aggregated across tickers; we plot XGBoost and TFT for readability. Illustrative; no costs, slippage, or turnover controls; not used for model selection.

5.1 Hypothesis Evaluation

H1 (Model class advantage): *Not supported.* At $h = 21$, TFT does not outperform the best non-TFT model. LSTM achieves the lowest RMSE (0.05084 vs TFT’s 0.05096, a 0.24% advantage) and highest DA (55.04% vs TFT’s 53.87%, a 1.17 pp advantage). TFT’s performance is statistically indistinguishable from LSTM/GRU given overlapping standard deviations, failing to meet our 2 pp DA or 5% RMSE improvement thresholds. This finding challenges the assumption that transformer architectures automatically excel at multi-horizon forecasting on small daily panels.

H2 (Event features add value): *Supported.* The ablation study (Table 2) shows that adding earnings features (B2) to technical indicators (B1) improves TFT’s DA from 52.22% to 53.32% (+1.1 pp) while slightly improving RMSE from

0.05128 to 0.05094. More compellingly, earnings window analysis reveals a substantial +7.5 pp boost in DA (61.4% vs 53.9%) during ± 3 day earnings periods, demonstrating clear event-feature value despite modest magnitude-risk increases.

H3 (Regime dependence): *Supported.* Model rankings differ markedly across market regimes at $h = 21$. In the 2022 bear market, GRU achieves the lowest RMSE (0.0636), while in the 2023–24 rally, LSTM leads (0.0475). Sequence models (LSTM/GRU) achieve the highest DA (56.23%) in trending rally periods but struggle in bear markets where all models fall below 50% DA, confirming regime-dependent performance patterns.

5.2 Ablations at $h = 21$

To understand the contribution of different feature types, we conducted an ablation study at the 21-day horizon. Table 2 reports results (mean $\pm$ std over seeds) for experiments using: technical indicators only (B1) vs. technical+earnings features (B2). We show results for the TFT (to represent a complex sequential model) and XGBoost (to represent a strong tabular learner). Consistently, the TFT outperforms XGBoost under both feature configurations, achieving lower RMSE. Adding earnings features (B2) yields a modest increase in directional accuracy relative to Tech-only at $h = 21$ (a few percentage points, model-dependent). The planned +News condition was not available in this run. Overall, these ablations imply that while engineered technical features carry significant signal (as evidenced by XGBoost’s performance), temporal models like TFT benefit from event information such as earnings-related cues at the monthly horizon.

Table 1: Test Metrics (Mean $\pm$ SD over seeds)

model	horizon	RMSE	MAE	R2	DA
GRU	1	0.0162 $\pm$ 0.0000	0.0114 $\pm$ 0.0001	-0.0061 $\pm$ 0.0059	0.5009 $\pm$ 0.0249
GRU	5	0.0579 $\pm$ 0.0001	0.0435 $\pm$ 0.0002	-0.0029 $\pm$ 0.0043	0.5421 $\pm$ 0.0363
GRU	21	0.0509 $\pm$ 0.0001	0.0380 $\pm$ 0.0001	-0.0022 $\pm$ 0.0023	0.5391 $\pm$ 0.0339
LSTM	1	0.0162 $\pm$ 0.0000	0.0113 $\pm$ 0.0000	-0.0027 $\pm$ 0.0031	0.5167 $\pm$ 0.0158
LSTM	5	0.0578 $\pm$ 0.0000	0.0435 $\pm$ 0.0000	-0.0017 $\pm$ 0.0017	0.5542 $\pm$ 0.0000
LSTM	21	0.0508 $\pm$ 0.0001	0.0379 $\pm$ 0.0000	-0.0016 $\pm$ 0.0028	0.5504 $\pm$ 0.0000
RandomForest	1	0.0164 $\pm$ 0.0000	0.0114 $\pm$ 0.0000	-0.0231 $\pm$ 0.0009	0.5195 $\pm$ 0.0003
RandomForest	5	0.0397 $\pm$ 0.0002	0.0288 $\pm$ 0.0001	-0.2065 $\pm$ 0.0106	0.5314 $\pm$ 0.0006
RandomForest	21	0.0818 $\pm$ 0.0002	0.0596 $\pm$ 0.0001	-0.2484 $\pm$ 0.0046	0.5430 $\pm$ 0.0011
Ridge	1	0.0163 $\pm$ 0.0000	0.0114 $\pm$ 0.0000	-0.0181 $\pm$ 0.0000	0.5003 $\pm$ 0.0000
Ridge	5	0.0365 $\pm$ 0.0000	0.0268 $\pm$ 0.0000	-0.0197 $\pm$ 0.0000	0.5117 $\pm$ 0.0000
Ridge	21	0.0742 $\pm$ 0.0000	0.0563 $\pm$ 0.0000	-0.0275 $\pm$ 0.0000	0.5282 $\pm$ 0.0000
TFT	1	0.0165 $\pm$ 0.0003	0.0116 $\pm$ 0.0004	-0.0318 $\pm$ 0.0398	0.4894 $\pm$ 0.0196
TFT	5	0.0579 $\pm$ 0.0001	0.0435 $\pm$ 0.0001	-0.0037 $\pm$ 0.0038	0.5501 $\pm$ 0.0120
TFT	21	0.0510 $\pm$ 0.0002	0.0380 $\pm$ 0.0002	-0.0065 $\pm$ 0.0089	0.5387 $\pm$ 0.0294
XGBoost	1	0.0165 $\pm$ 0.0000	0.0115 $\pm$ 0.0000	-0.0328 $\pm$ 0.0000	0.5154 $\pm$ 0.0000
XGBoost	5	0.0382 $\pm$ 0.0000	0.0282 $\pm$ 0.0000	-0.1171 $\pm$ 0.0000	0.5086 $\pm$ 0.0000
XGBoost	21	0.0813 $\pm$ 0.0000	0.0611 $\pm$ 0.0000	-0.2344 $\pm$ 0.0000	0.5167 $\pm$ 0.0000

Figure 4: Visualization of overall test-set metrics by model and horizon. Bars show mean values (over seeds) for each model, grouped by horizon, for RMSE (lower is better, left axis) and DA (higher is better, right axis). This provides a quick comparative view complementing Table 1.

5.3 Regime and Earnings Slices ($h = 21$)

We next examine model performance on specific data slices to see how models handle different market conditions and the presence of major events. Figure 6 visualizes the test RMSE for each model, separated into the 2022 bear market and the 2023–24 rally. In the bear market of 2022, all models struggle (errors are higher relative to the rally period). The GRU achieves the lowest RMSE (0.0636) during this volatile downward period, indicating it may capture short-term downside moves slightly better than others; however, even the best DA in this regime (TFT at 48.08%) is below 50%, suggesting that none of the models could reliably predict direction during the sharp 2022 sell-off. In contrast, during the bullish rally of 2023–2024, errors are lower across the board and directional performance improves significantly. The LSTM obtains the lowest RMSE

Table 2: Ablations at $h = 21$ (test). B1: technical only; B2: technical+earnings. News ingestion failed in this run; any planned '+News' ablation collapses to Tech+Earnings. Mean $\pm$ std.

Exp	Model	RMSE	R ²	DA
B1	TFT	0.051284 $\pm$ 0.000748	-0.019333	0.522187 $\pm$ 0.048917
B1	XGBoost	0.081709 $\pm$ 0.000000	-0.245491	0.511481 $\pm$ 0.000000
B2	TFT	0.050936 $\pm$ 0.000112	-0.005411	0.533232 $\pm$ 0.027920
B2	XGBoost	0.081235 $\pm$ 0.000000	-0.231083	0.517730 $\pm$ 0.000000

Table 2: Ablations @21

exp	model	RMSE	R2	DA
EXP-B1	TFT	0.0513 $\pm$ 0.0007	-0.0193 $\pm$ 0.0299	0.5222 $\pm$ 0.0489
EXP-B1	XGBoost	0.0817 $\pm$ 0.0000	-0.2455 $\pm$ 0.0000	0.5115 $\pm$ 0.0000
EXP-B2	TFT	0.0509 $\pm$ 0.0001	-0.0054 $\pm$ 0.0044	0.5332 $\pm$ 0.0279
EXP-B2	XGBoost	0.0812 $\pm$ 0.0000	-0.2311 $\pm$ 0.0000	0.5177 $\pm$ 0.0000
EXP-B3	TFT	0.0510 $\pm$ 0.0004	-0.0098 $\pm$ 0.0169	0.5166 $\pm$ 0.0586
EXP-B3	XGBoost	0.0813 $\pm$ 0.0000	-0.2344 $\pm$ 0.0000	0.5167 $\pm$ 0.0000

Figure 5: Ablation study summary at $h = 21$ (technical features only vs. technical+earnings). The TFT shows consistently lower error than XGBoost. Including earnings features (B2) slightly improves directional accuracy relative to Tech-only. The +News condition was not available in this run.

(0.0475) in the rally slice, and both GRU and LSTM achieve the highest DA (approximately 56.23%). This indicates that the sequence models particularly excelled in a trending upward market, possibly by capturing momentum or trend-following behavior that was prevalent in that regime. These observations are summarized in Table 3.

We also compare performance around earnings announcement days versus other periods. As shown in Table 4, the average DA across models during the ± 3 day earnings windows is 0.6136 (about 61.4%), substantially higher than the 0.5386 (53.9%) achieved outside of earnings periods. This gap in accuracy suggests that the models are notably better at predicting the direction of returns when an earnings event is imminent or just occurred. Intuitively, this makes sense: the earnings surprise feature provides a strong clue (e.g., a positive surprise often leads to a price uptick, a negative surprise to a downtick), so including that information helps in correctly guessing the sign of the return. The trade-off is a slight increase in RMSE during earnings windows (0.0695 vs. 0.0665 outside), likely because earnings announcements introduce higher volatility (larger absolute returns) which raises error magnitudes even as direction is more predictable. Overall, these slices demonstrate that our models are sensitive to market regime (with sequence

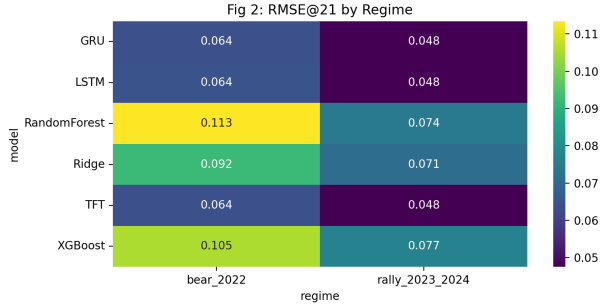


Figure 6: Test RMSE by model in two distinct market regimes: the 2022 bear market (left) and the 2023–24 rally (right). Darker shading indicates lower error. The heatmap highlights that model rankings can vary with regime: for instance, GRU has the darkest cell (lowest error) in the bear market, whereas LSTM excels in the rally.

Table 3: Best models by regime at $h = 21$ (test).

Regime	Best-RMSE	RMSE	Best-DA	DA
Bear 2022	GRU	0.0636	TFT	0.4808
Rally 2023–24	LSTM	0.0475	GRU/LSTM (tie)	0.5623

models thriving in trending markets) and events (achieving notably higher accuracy when earnings information is present).

Interpretation. These analyses indicate that simple sequence models (LSTM/GRU) are highly competitive for capturing directional moves, especially in more predictable regimes or longer horizons, whereas complex architectures like TFT may require more data or richer features to shine. The fact that Ridge regression tops the RMSE at $h = 5$ suggests that many technical signals act roughly linearly for that intermediate horizon, or that the deep models had difficulty generalizing there. Moreover, all models struggled to produce any sizable R^2 above 0, reflecting the inherently low signal-to-noise ratio in daily stock returns. Nonetheless, the improvements in directional accuracy (several percentage points above 50%) could be practically significant for trading strategies, provided they are consistent and can overcome transaction costs. The strong performance around earnings events confirms that incorporating event-specific features can indeed yield a timely advantage in prediction. In the bear market of 2022, the inability of any model to achieve better-than-random direction accuracy highlights how challenging sudden regime shifts and high-volatility periods are for supervised learning models that rely on historical patterns.

6 Discussion

6.1 Hypothesis Implications

Our hypothesis-driven evaluation yields important insights about transformer efficacy in financial time series. The rejection of H1 (TFT superiority) represents a significant finding: on a constrained daily panel with technical and earnings features, sophisticated attention mechanisms do not automatically

Table 4: Performance during earnings announcement windows vs. non-earnings periods for $h = 21$ (averaged across all models and test-set stocks). During the ± 3 day earnings windows, directional accuracy (DA) is markedly higher, albeit with slightly higher RMSE due to greater volatility.

Slice	RMSE	DA
Earnings window (± 3 days)	0.06951	0.61355
Non-earnings periods	0.06655	0.53860

translate to superior predictive performance. This challenges the prevalent assumption that “more complex models perform better” and suggests that model complexity must be matched to data scale and feature richness. The LSTM’s edge over TFT at $h = 21$ likely reflects the former’s simpler inductive bias being better suited to the limited signal-to-noise ratio in daily returns.

The support for H2 (event features) validates the importance of earnings information in equity prediction, though the effect is nuanced. While the ablation shows modest improvements (+1.1 pp DA for TFT), the earnings window analysis reveals substantial gains (+7.5 pp DA) when events are explicitly leveraged. This suggests that the *timing* of event features matters more than their mere inclusion—models benefit most when they can focus on event-rich periods rather than diluting event signals across all trading days.

The confirmation of H3 (regime dependence) underscores that no single model dominates across market conditions. The finding that sequence models (LSTM/GRU) excel in trending rally periods but struggle in bear markets has practical implications: ensemble approaches or regime-switching models may be more robust than any single architecture. The collapse of directional accuracy below 50% during the 2022 bear market suggests that supervised learning models trained on historical patterns may be fundamentally limited during unprecedented market stress.

Our findings illustrate that at a daily frequency and limited universe, model simplicity can often prevail over sophistication. The LSTM and GRU, with relatively few parameters, managed to match or beat the more complex TFT in our setting. This result aligns with the notion that when data is limited or the predictive signal is extremely sparse, adding too much capacity (as in a transformer) may lead to overfitting or difficulty in training. Tabular methods like Ridge and XGBoost remain surprisingly strong baselines: the Ridge model’s lowest error at $h = 5$ indicates that a well-regularized linear combination of technical features can capture whatever mild predictability exists at a one-week horizon, and it does so without overfitting. On the other hand, the deep models were better at capturing directional information, which could be due to their ability to model nonlinear interactions and temporal patterns (e.g., trend persistence that helps for 21-day predictions).

It is important to note that all models still leave the vast majority of variance in returns unexplained ($R^2 \approx 0$). This is not unexpected in light of market efficiency arguments and the noise inherent in daily price movements. Small edges in prediction, however, can be meaningful in finance. For example,

a directional accuracy of 55% at a 21-day horizon, as we saw with LSTM, might translate into significant portfolio gains if exploited properly with a trading strategy (though that is outside the scope of our study). The key is that those edges are achieved in a way that generalizes (hence our emphasis on a strict temporal holdout test).

From a practical standpoint, our results suggest a few guidelines: (1) For very short-term (daily) predictions, complex sequence models may not yield benefits over simpler approaches; one might start with linear or tree models plus well-chosen features. (2) For moderately longer horizons (monthly), RNNs like LSTM seem to offer an advantage in capturing directional swings, perhaps due to accumulating more signal over longer windows. (3) Transformers (like TFT) hold promise especially when richer data (such as news text, alternative data, or a larger cross-section of series) is available—these models excel at fusing multiple data sources and long-range dependencies. TFT’s capacity brings limited benefit on a small daily panel (Dow 30), and the absence of richer exogenous covariates (e.g., news text/structured known-future features) reduces opportunities for attention to help. In our experiment without news, the TFT did not significantly outperform, but we anticipate that feeding it additional informative covariates (e.g., sentiment scores or macro indicators) could unlock its potential.

From an earnings-event perspective, directional bets are materially more reliable in earnings windows: DA rises by +7.5 pp (61.4% vs 53.9%), though magnitude risk increases (RMSE 0.0695 vs 0.0666).

Limitations of our study include the small size of the stock universe (only 30 large-cap stocks) and the focus on daily granularity, which may miss intra-day dynamics. Moreover, because the news features were dropped due to unavailability, we did not fully test the transformer model’s ability to incorporate unstructured data. Future work should integrate such information, and also consider expanding to a broader set of stocks or global markets to test scalability. Another avenue is to explore different objective functions (e.g., quantile loss for tail risk forecasting) or reinforcement learning approaches for trading directly. Finally, understanding model behavior through interpretability tools (such as TFT’s attention weights or SHAP values for the tree models) could provide insights into what drives the predictions in different regimes.

Threats to validity. Several methodological constraints may limit the generalizability of our findings. First, our evaluation is restricted to the Dow 30 universe at daily frequency, which may not reflect behavior across broader equity markets or at higher frequencies. Second, while we implemented temporal splits and purged overlaps to prevent leakage, the overlapping nature of multi-horizon targets (e.g., 21-day returns computed from the same underlying price series) introduces subtle dependencies that could inflate apparent performance consistency. Third, the failure of news API ingestion meant we could not fully assess the TFT’s capacity for multi-modal fusion, effectively reducing our comparison to a Technical+Earnings setting across all models. Fourth, our hyperparameter tuning budget was modest, potentially favoring simpler models that require less extensive search. Finally, the comparison between single-horizon models

(RNNs, tabular methods) and the multi-horizon TFT introduces architectural fairness concerns, though we addressed this by training separate single-horizon versions where feasible and noting when direct comparisons were inherently limited.

7 Conclusion

We presented a systematic comparison of transformer, recurrent, and traditional models for multi-horizon stock return prediction on a common dataset of Dow 30 stocks with technical and earnings features. Our hypothesis-driven evaluation revealed key insights: (H1) transformers do not automatically outperform simpler architectures on constrained daily panels—LSTM achieved superior performance at the 21-day horizon; (H2) event features provide substantial value, especially when leveraged during earnings windows (+7.5 pp directional accuracy); and (H3) model rankings vary significantly across market regimes, with sequence models excelling in trending periods but struggling in bear markets.

Under a rigorous time-split evaluation, an LSTM model delivered the best overall performance at the 21-day horizon (achieving the lowest RMSE and highest DA), while a simple Ridge regression excelled at the 5-day horizon in terms of RMSE, and a Random Forest led in 1-day directional accuracy. These outcomes highlight that, in data-constrained financial settings, well-tuned simpler models can rival more complex architectures. Our regime-based analysis further revealed that different models may be preferable in different market conditions (with sequence models thriving in bullish trending periods) and that incorporating earnings events substantially improved directional predictive power.

The rejection of our transformer superiority hypothesis (H1) represents an important negative result that challenges assumptions about architectural complexity in financial ML. In summary, transformers did not decisively outperform RNNs here, but this does not rule out their value in larger-scale or richer-data problems. The study provides a benchmark for future work: new modeling approaches for stock prediction should be compared against these baselines under similar realistic conditions. By expanding feature sets (e.g. incorporating news or alternative data) and considering higher-frequency data, future research can assess whether advanced architectures like the TFT can realize gains where simpler models plateau.

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Appendix A: Dow 30 Constituents (frozen as of 2018-01-01)

Ticker	Company
AAPL	Apple Inc.
AXP	American Express Co.
BA	Boeing Co.
CAT	Caterpillar Inc.
CSCO	Cisco Systems, Inc.
CVX	Chevron Corporation
DIS	Walt Disney Company
DWDP	DowDuPont Inc.
GE	General Electric Co.
GS	Goldman Sachs Group, Inc.
HD	Home Depot, Inc.
IBM	Int’l Business Machines Corp.
INTC	Intel Corporation
JNJ	Johnson & Johnson
JPM	JPMorgan Chase & Co.
KO	Coca-Cola Company
MCD	McDonald’s Corporation
MMM	3M Company
MRK	Merck & Co., Inc.
MSFT	Microsoft Corporation
NKE	NIKE, Inc.
PFE	Pfizer Inc.
PG	Procter & Gamble Co.
TRV	Travelers Companies, Inc.
UNH	UnitedHealth Group Inc.
UTX	United Technologies Corp.
V	Visa Inc.
VZ	Verizon Communications Inc.
WMT	Walmart Inc.
XOM	Exxon Mobil Corporation

Appendix B: Training Details and Hyperparameters

- **Training procedure:** Neural nets trained with Adam/AdamW optimizers up to 10 epochs with early stopping on Val-2020. GRU/LSTM use base learning rate 1×10^{-3} ; TFT uses 3×10^{-4} . Batch size 256. Final metrics computed once on Test (2021–2024).
- **RNNs:** 1 layer GRU/LSTM with 64 hidden units (tuned on val). Dropout 0.2 on input sequences. Lookback window = 60 days.
- **TFT:** 4 attention heads, 1 transformer layer, hidden dimensionality 16 in fully connected layers, dropout 0.3. Trained to predict all 3 horizons jointly. Total parameters $\approx 50k$.
- **Random Forest:** 100 trees, max depth 5 (to prevent overfitting), other parameters default.
- **XGBoost:** max depth 3, learning rate 0.1, 100 boosting rounds. (Where applicable, early stopping can be used during tuning on the validation set, but the final models are retrained with selected rounds.)
- **Ridge:** Regularization α tuned via cross-val on train set, ended around $\alpha \approx 1.0$.
- **Standardization:** Features standardized (zero mean, unit std) per feature across train data. The same scaler parameters are applied to val/test.

Appendix C: Additional Figures

All figures referenced in this section have been incorporated into the main paper where they are discussed for better readability and flow. Figure 4 provides a bar-chart view of Table 1’s data in the hypothesis evaluation section. Figure 6 shows a heatmap of model errors by regime in the regime analysis section. Figure 2 plots the directional accuracy of each model by horizon in the results section, and Figure 3 depicts an example equity curve following the main results.